

Bayesian Model Averaging in Meta-Analysis

A Simple Application

Tomas Havranek

Czech National Bank
Charles University in Prague

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What Is BMA?

- A method used to deal with model uncertainty
- Many explanatory variables → problems with model selection
- Sequential t-tests problematic (each conditional on the previous one)
- BMA runs many regressions with different subsets of the explanatory variables
- Weighted by goodness of fit and model complexity

How Does It Work in Practice?

- We can estimate many OLS regressions with different subsets and weigh them by adjusted R^2
- Problem: usually billions of subsets
- Bayesian methods make computation easier
- Monte Carlo Markov Chain Algorithm typically used
- Estimated in R or Matlab using the BMS package

Application: Elasticity of Intertemporal Substitution

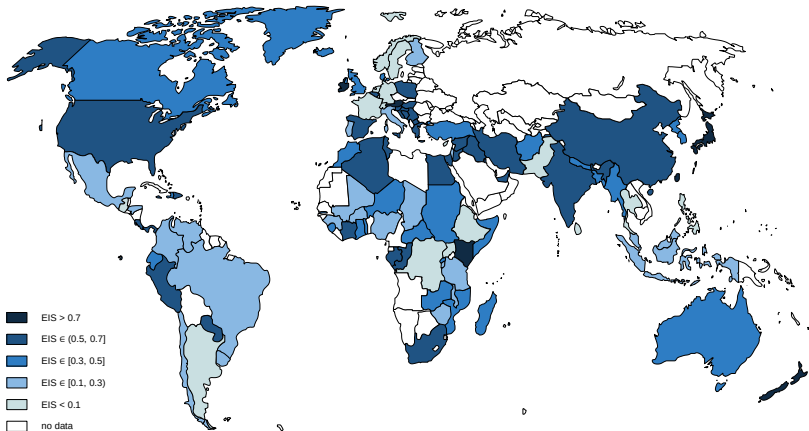
The EIS reflects households' willingness to substitute consumption between time periods in response to changes in the expected real interest rate.

$$u(c) = \frac{c^{1-\frac{1}{EIS}} - 1}{1 - \frac{1}{EIS}}; \quad \text{if } EIS = 1 \Rightarrow u(c) = \log c.$$

Crucial in models involving intertemporal choice:

- monetary policy,
- fiscal policy,
- portfolio choice,
- computing the social cost of carbon emissions, and more.

The Elasticity Varies Across Countries.



What Explains the Differences?

Country-level variables

Stock market participation: Euler equation valid for asset holders.

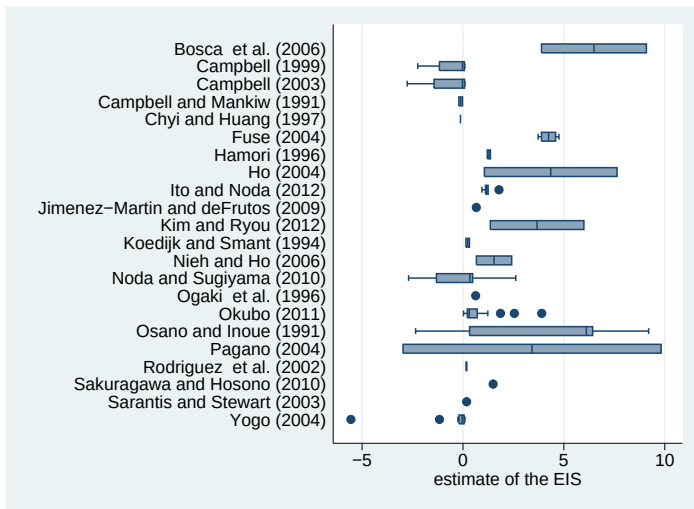
GDP per capita: necessities hard to substitute across time periods.

Credit availability: financial constraints hamper intertemporal substitution.

Real interest: the elasticity does not have to be constant.

Rule of law: institutions can affect financial decisions.

The Elasticity Varies Across Methods.



Variables Coded (1)

Utility

Epstein-Zin	=1 if the estimation differentiates between the EIS and the coefficient of relative risk aversion.
Habits	=1 if habits in consumption are assumed.
Nonsep. durables	=1 if the model allows for nonseparability between durables and nondurables.
Nonsep. public	=1 if the model allows for nonseparability between private and public consumption.
Nonsep. trad-ables	=1 if the model allows for nonseparability between tradables and nontradables.

Data

No. of households	The logarithm of the number of cross-sectional units used in the estimation (households, cohorts, countries).
No. of years	The logarithm of the number of years of the data period used in the estimation.
Average year	The logarithm of the average year of the data period.
Micro data	=1 if the coefficient comes from a micro-level estimation.
Annual data	=1 if the data frequency is annual.
Monthly data	=1 if the data frequency is monthly.

Variables Coded (2)

Design

Quasipanel	=1 if quasipanel (synthetic cohort) data are used.
Inverse estimation	=1 if the rate of return is the dependent variable in the estimation.
Asset holders	=1 if the estimate is related to the rich or asset holders.
First lag instrument	=1 if the first lags of variables are included among instruments.
No year dummies	=1 if year dummies are omitted in micro studies using the Panel Study of Income Dynamics.
Income	=1 if income is included in the specification.
Taste shifters	The logarithm of the number of controls for taste shifters.

Variable definition

Total consumption	=1 if total consumption is used in the estimation.
Food	=1 if food is used as a proxy for nondurables.
Stock return	=1 if the rate of return is measured as stock return.
Capital return	=1 if the rate of return is measured as the return on capital.

Variables Coded (3)

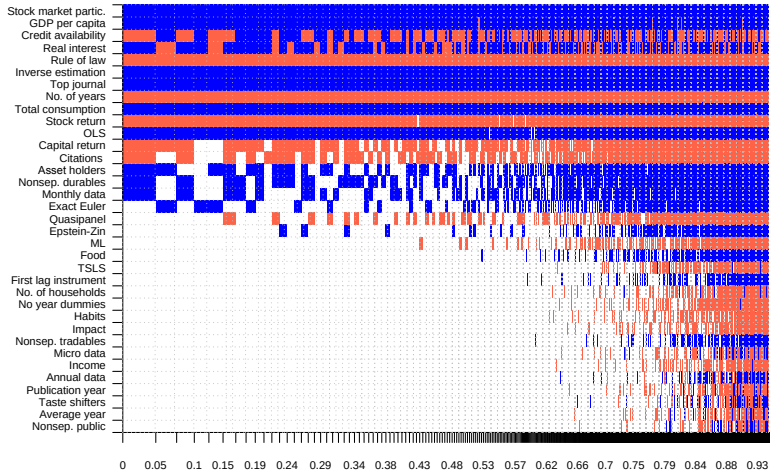
Estimation

Exact Euler	=1 if the exact Euler equation is estimated.
ML	=1 if maximum likelihood methods are used for estimation.
TSLS	=1 if two-stage least squares are used for estimation.
OLS	=1 if ordinary least squares are used for estimation.

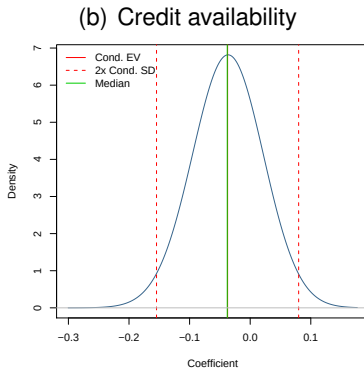
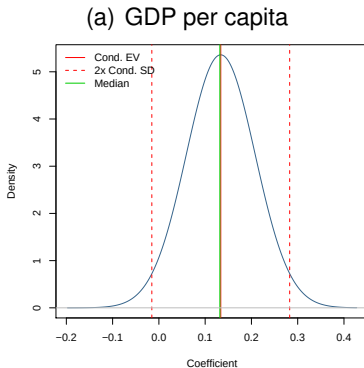
Publication

SE	The reported standard error of the estimate of the EIS.
Publication year	The logarithm of the year of publication of the study.
Citations	The logarithm of the number of per-year citations of the study in Google Scholar.
Top journal	=1 if the study was published in one of the top five journals in economics.
Impact	The recursive RePEc impact factor of the outlet.

Bayesian Model Averaging

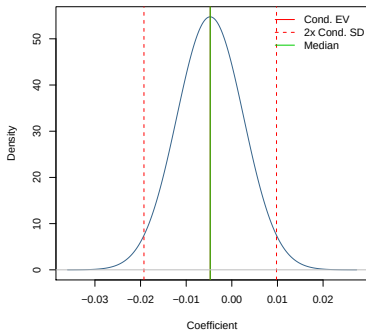


Posterior Coefficient Distributions (1)

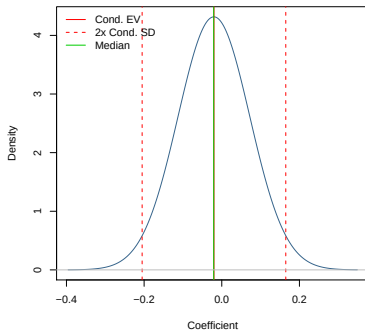


Posterior Coefficient Distributions (2)

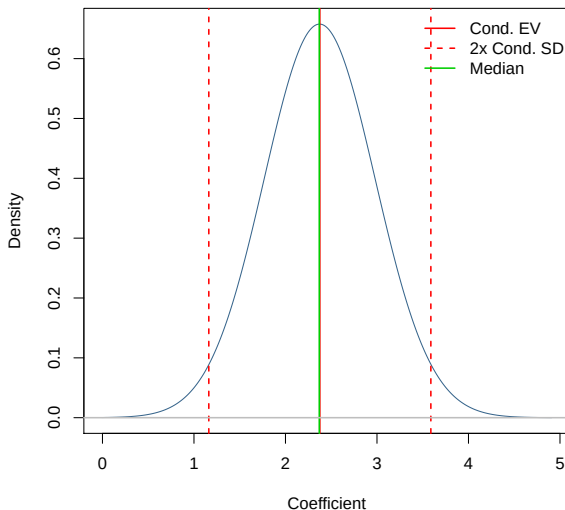
(c) Real interest



(d) Rule of law



Stock Market Participation



Economic Significance

Stock market participation and GDP per capita affect the elasticity a lot:

Variable	Maximum effect	Std. dev. effect
Stock market partic.	0.931	0.141
GDP per capita	0.683	0.088
Credit availability	-0.119	-0.020
Real interest	-0.265	-0.019
Rule of law	-0.087	-0.012

Summary

In a Nutshell. . .

- 1 Bayesian model averaging is useful in regressions with many explanatory variables.
- 2 It can be thought of as a generalization of the typical practice of conducting robustness checks with different sets of explanatory variables.
- 3 BMA is easy to use!

Project Website

www.meta-analysis.cz

For Further Reading

-  Koop, G. (2003): Bayesian Econometrics.
Wiley, 1st. edition.
-  Havranek, T., M. Rusnak (2013): Transmission Lags of Monetary Policy: A Meta-Analysis.
International Journal of Central Banking: **9(4)**: pp. 39–76.
-  Havranek, T., R. Horvath, Z. Irsova, & M. Rusnak (2015): Cross-Country Heterogeneity in Intertemporal Substitution.
Journal of International Economics: **96(1)**: pp. 100–118.

Reading list on RePEc: Google “meta-analysis in economics.”